MDP

Customer Churn Management

01

STATE SPACE S

9 states = Tenure × Recency

New | Mid | Loyal

Active | Neutral | Inactive

e.g. **Mid_Inactive** — mid-tenure, >7d → high risk

02

ACTION A

Binary: coupon or not

A=0 · Do Nothing
Cost = $0

A=1 · Send Coupon
Cost = $20

03

TRANSITION T(s'|s,a)

From **5,630** records · Each state → Stay or Churn

Mid_Inactive example:

No coupon → Churn **60%**

With coupon → Churn **25%**

35pp reduction = biggest coupon impact

04

REWARD R(s,a) & γ

```
R = P(stay)×+$100
    + P(churn)×-$100
    - coupon cost
```

- Retain: +$100
- Churn: -$100
- Coupon: -$20

γ = 0.9 Future rewards discounted

Churn Analysis — Where Coupons Matter

Churn Rate (No Coupon)

	Active	Neutral	Inactive
New	47%	35%	37%
Mid	30%	40%	60%
Loyal	11%	8%	9%

Churn Rate (With Coupon)

	Active	Neutral	Inactive
New	44%	32%	35%
Mid	18%	22%	25%
Loyal	10%	7%	8%

Coupon Lift (Δ Churn)

	Active	Neutral	Inactive
New	3%	3%	2%
Mid	12%	18%	35%
Loyal	1%	1%	1%

Mid_Inactive: Biggest Opportunity

60% → 25% churn with coupon. A 35pp reduction — the largest lift across all 9 states. This is why the algorithm targets it.

Loyal & New: Coupon Doesn't Help

Loyal: already <11% churn, lift is only 1–2pp. **New:** high churn but lift is only 2–5pp — not enough to justify $20.

Policy Iteration — How It Works

Step 1: Policy Evaluation

Bellman Equation:

$$V(s) = R(s,a) + \gamma \cdot \Sigma\, P(s'|s,a) \cdot V(s')$$

(immediate reward + discounted future)

Calculate long-term value of each state.
Repeat until values converge ($\Delta < \theta$).

→

Step 2: Policy Improvement

Greedy selection:

$$\pi(s) = \text{argmax_a}\ Q(s,a)$$

(pick action with highest long-term value)

Compare Q-values: coupon vs no coupon.
If any action changes → back to Step 1.

Converged in 2 iterations — Iter 1: Mid_Inactive & Mid_Neutral → coupon | Iter 2: stable, done

Policy Iteration — Results

78%↓
Coupon Spend

2.3×↑
Profit/User

2/9
Targeted

USER STATE	ACTION	LTV	STRATEGIC INSIGHT
Loyal_Active	None	$727.72	Already loyal — no intervention
Loyal_Inactive	None	$782.89	Retains naturally
Loyal_Neutral	None	$820.65	Highest long-term value
Mid_Active	None	$258.87	Monitor — not at risk yet
Mid_Inactive	**Coupon ✓**	**$565.95**	**Primary target — high churn risk**
Mid_Neutral	**Coupon ✓**	**$419.91**	**Primary target — high lift**
New_Active	None	$56.46	Low LTV — not cost-effective
New_Inactive	None	$161.02	Early lifecycle
New_Neutral	None	$187.07	Early lifecycle

Focus on Mid-tier

$20 coupon protects $300+ LTV. Only states with positive ROI.

Skip Loyalists

Already retain. $20 wasted, no retention gain.

New = Not Yet

LTV too low ($56–187). Let them mature first.